Adiposity drives rheumatoid arthritis through distinct mechanisms in younger patients

Tyler Hatstadt[1], Michael Bryan[2]

[1] Rye Country Day School, Rye, United States; [2] University of Oxford, Oxford, United Kingdom

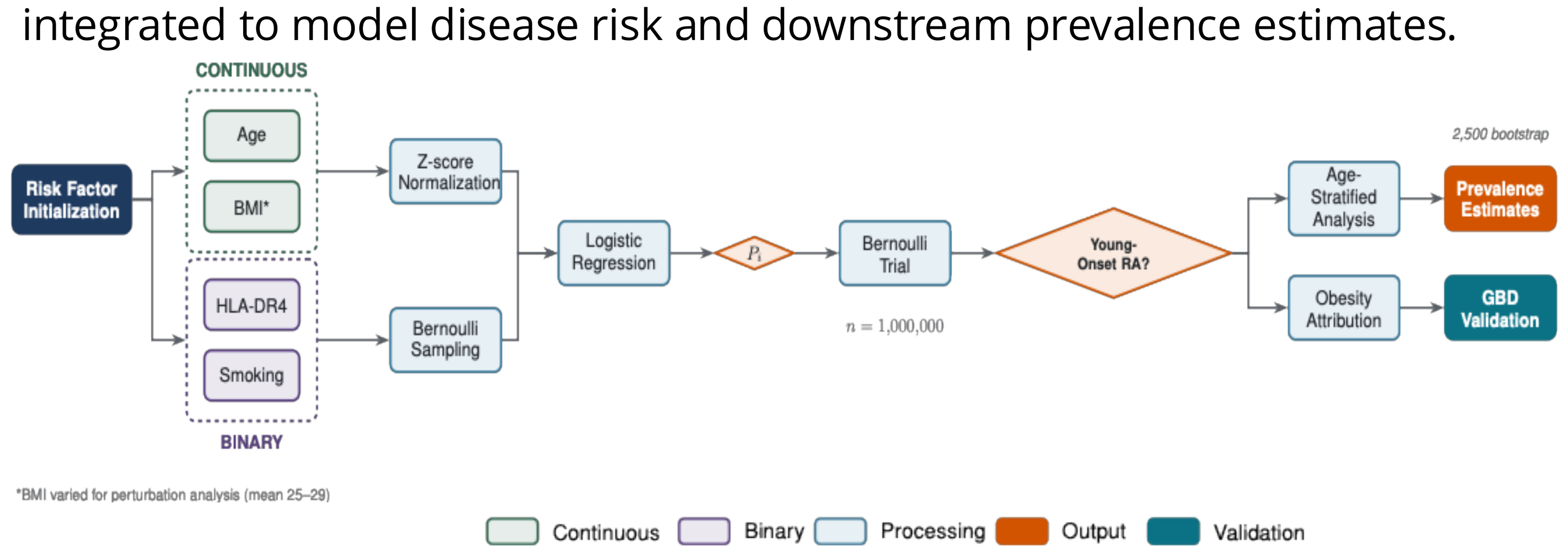
Introduction

- Incidence of rheumatoid and juvenile arthritis (RA/JA) in individuals under 40 has risen steadily since 1990, in parallel with the global increase in childhood obesity.
- Although obesity is known to promote systemic inflammation, its role in RA pathogenesis remains poorly defined, and adiposity is typically treated as a comorbidity and amplifier rather than a potential upstream driver.
- RA/JA provides a cleaner test of the adiposity hypothesis than adult RA, as classical risk factors largely do not apply - children do not smoke, and the HLA-DR4 shared epitope plays a limited role in most subtypes.
- We hypothesized that childhood obesity contributes to young-onset RA through adipokine-mediated inflammatory pathways, representing a distinct etiological mechanism from older-onset disease.
- Because conventional epidemiological approaches struggle to capture the long latency between childhood exposures and autoimmune disease, we developed a simulation model to estimate the proportion of the rise in young-onset arthritis attributable to childhood obesity.

Methods

- We constructed a Monte Carlo simulation incorporating published odds ratios for established RA risk factors, including obesity, smoking, and HLA-DR4 positivity, along with their multiplicative interactions.
- The model followed 1,000,000 simulated individuals, integrating age-specific prevalences for binary risk factors and appropriate distributions for continuous variables.
- Sensitivity analyses focused on the exponential decay function used to model age-dependent BMI effects, applied to 100,000 simulated individuals.
- Model outputs were calibrated against Global Burden of Disease prevalence estimates, and 2,500 iterations were run to propagate parameter uncertainty.
- Perturbation analyses varied mean BMI from 25 to 29 while holding other risk factors constant.

Figure 1. A novel simulation framework enables estimation of young-onset RA prevalence and obesity attribution. The schematic summarizes how age, BMI, genetic, and environmental risk factors are integrated to model disease risk and downstream prevalence estimates.



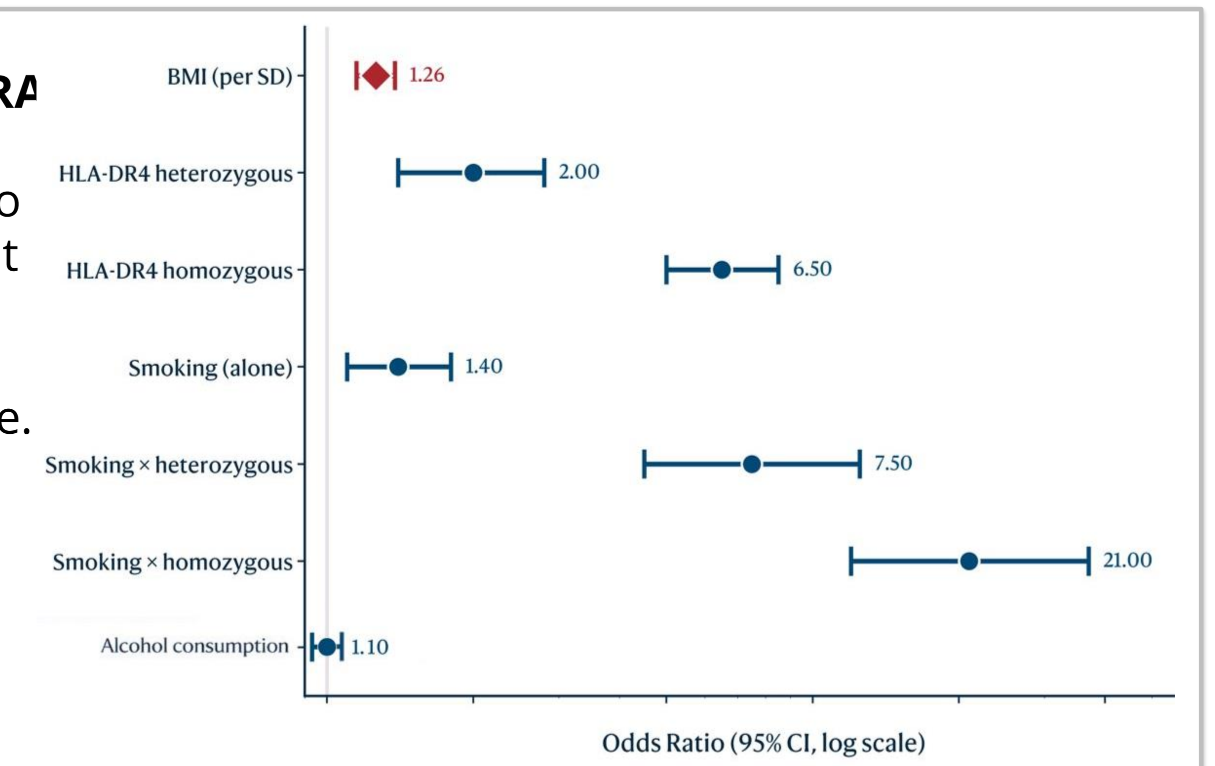
Key Takeaways

1. Using a Monte Carlo simulation, we systematically perturbed average BMI in younger populations to quantify its contribution to rising arthritis prevalence.
2. Each unit increase in mean BMI was associated with an average 0.005% increase in arthritis prevalence.
3. The disproportionate contribution of BMI in younger people suggests increased arthritis risk may be an under-recognized consequence of the ongoing childhood obesity epidemic.

Contact: thatstadt@gmail.com, michael.bryan@new.ox.ac.uk

Results

Figure 2. Observational evidence supports an association between elevated BMI and RA risk, distinct from genetic and smoking effects. Genotypic frequencies are assumed to remain stable over time, whereas BMI varies at the population level. Although BMI has the smallest effect size among the risk factors shown, it is the most difficult to control at scale. Smoking is unlikely to explain rising young-onset RA incidence given declining prevalence in younger populations. Alcohol consumption showed no statistically significant association and was therefore excluded from the simulation.



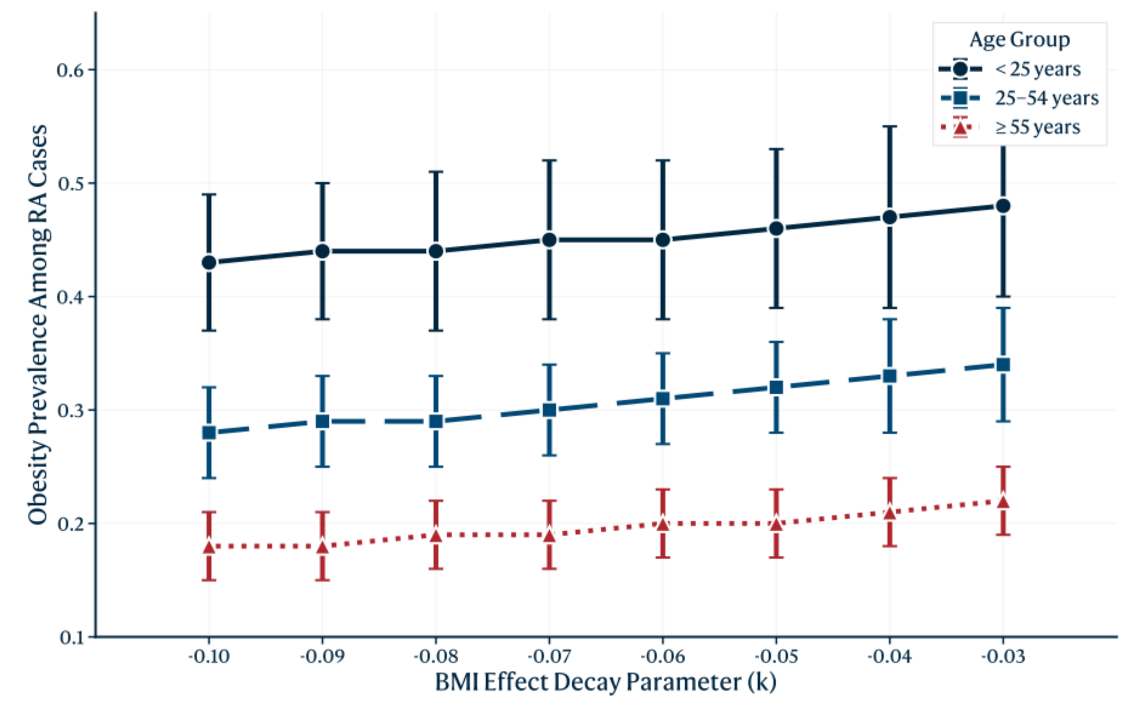


Figure 3. Model predictions are robust to assumptions about age-dependent attenuation of BMI effects (k). Sensitivity analyses varying the exponential decay constant showed negligible impact on estimated obesity prevalence among individuals with arthritis.

Figure 4. Counterfactual modelling suggests childhood obesity materially contributes to rising young-onset RA prevalence. Holding other risk factors constant, perturbation analyses identified a locally linear relationship between mean BMI and arthritis prevalence, with a predicted ~30% reduction in BMI-attributable diagnoses if average BMI returned to 1990 levels (<26).

Avg BMI scenario	Arthritis prevalence	Mean BMI	Obesity prevalence
25	0.05% (CI: 0.05–0.06%)	25.94 (CI: 25.19–26.69)	22.37% (CI: 16.33–29.14%)
26	0.06% (CI: 0.05–0.06%)	26.86 (CI: 26.13–27.60)	29.68% (CI: 22.96–37.28%)
27	0.06% (CI: 0.05–0.07%)	27.88 (CI: 27.17–28.59)	34.38% (CI: 27.78–41.57%)
28	0.06% (CI: 0.05–0.07%)	28.90 (CI: 28.21–29.63)	43.46% (CI: 36.47–50.90%)
29	0.07% (CI: 0.06–0.08%)	29.88 (CI: 29.21–30.60)	53.06% (CI: 45.95–60.31%)

Figure 5. Simulation outputs closely match Global Burden of Disease prevalence estimates, supporting external validity. Overlapping confidence intervals across age strata and the large simulated cohort (n = 1,000,000) indicate limited systematic error.

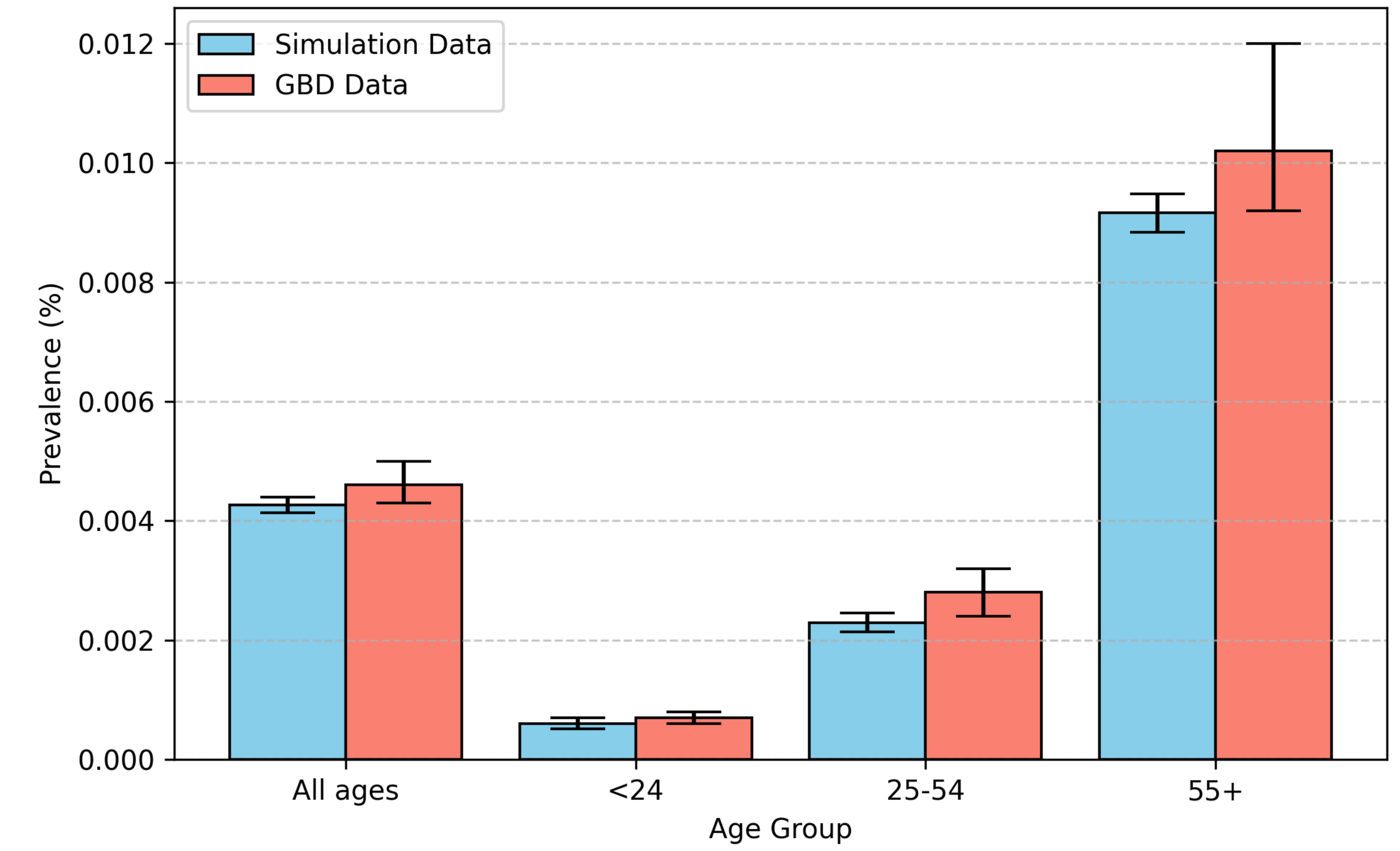
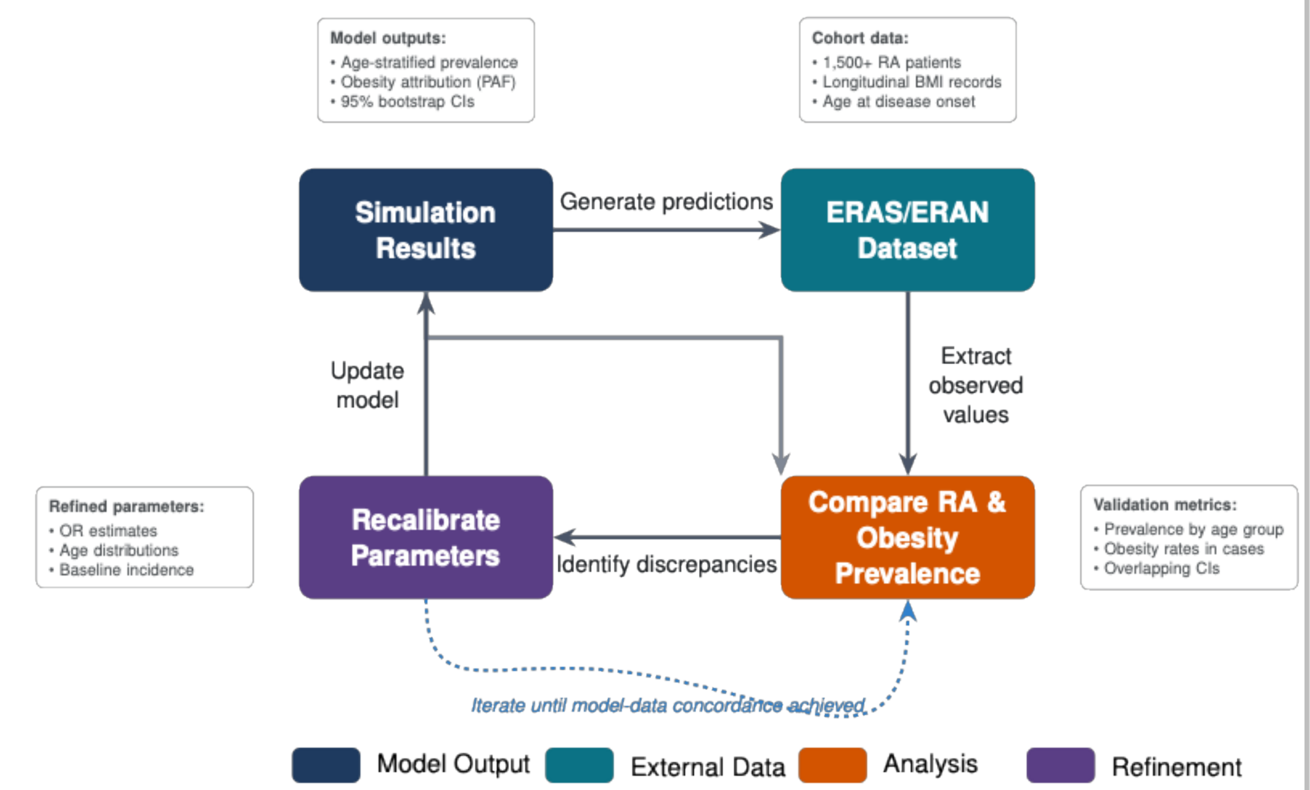


Figure 6. Prospective cohort validation will test model predictions and strengthen clinical relevance. Planned analyses using the ERAS and ERAN cohorts (King's College, London) will refine model parameters and support translation alongside population-level validation using GBD data.



Conclusions

- Our simulation modelling identifies childhood obesity as a potentially modifiable driver of young-onset RA/JA, suggesting that adiposity may act through mechanisms distinct from those in older patients.
- These findings provide a quantitative framework for incorporating RA prevention into the public health case for childhood obesity interventions.
- While simulation cannot establish causality, this approach allows integration of multiple risk factors across long time horizons and enables counterfactual analyses that are not feasible in observational studies.
- Introducing an explicit lag period between exposure and disease onset may further improve future models by capturing incidence and ageing effects within age strata.
- Prospective validation in longitudinal cohorts, together with mechanistic studies of adipokine-mediated inflammation, will be required to confirm these estimates and inform targeted prevention strategies.